

FRONTEND PROJECT DOCUMENTATION

Project Title: Smart Hotel & Restaurant Digital Management System (Frontend)

Type: Mobile-first SaaS Web Application

Primary Users: Hotel Owner, Waiter/Cashier, Kitchen Staff, Platform Admin

1. Project Overview

This project is the frontend application for a service-based SaaS platform designed for hotels and restaurants to digitally manage orders, billing, kitchen communication, and business reporting — using only mobile devices, with no paper bills or printers. The goal is to provide a simple, fast, modern, and user-friendly interface adaptable to small and medium-scale food businesses.

2. Frontend Objectives

- Provide a fully responsive mobile-first UI
- Deliver clean, modern, and futuristic user experience
- Ensure UI is simple for non-technical hotel staff
- Implement role-based access UI screens
- Ensure smooth state management and component-based architecture
- Build scalable, reusable UI components
- Integrate with backend via REST/GraphQL APIs (future stage)

3. Core Features (Frontend Scope)

- Futuristic login page (animated + food-theme UI)
- Role-based UI
- Mobile-first layouts
- Order management UI
- Kitchen live display screen
- Digital billing UI
- Analytics dashboards
- Offline handling (future)

4. Frontend Tech Stack

Framework: React + TypeScript

Bundler: Vite

Styling: Tailwind CSS

Animations: Framer Motion

Icons: lucide-react / custom SVG

Fonts: Google Fonts / Custom WebFonts

State Management: Zustand / Context API / Redux (later decision)

Form Handling: React Hook Form (planned)

Deployment: Vercel / Netlify / Firebase Hosting

5. Project Structure (Proposed)

src/

■■■ assets/

■■■ components/

■ ■■■■ buttons/

■ ■■■■ inputs/

- ■■■ cards/
- layouts/
- pages/
- ■■■ auth/
- ■■■ LoginPage.tsx
- context/
- hooks/
- animations/
- services/
- types/
- utils/
- main.tsx

6. UI/UX Principles

- Mobile-first strategy
- Minimal clicks, intuitive navigation
- Large touch targets
- Visual hierarchy for rapid reading
- Multi-language potential (English/Tamil/Hindi)
- Food, kitchen, and hotel themed identity

7. Component Architecture Plan

- Global reusable components
- Page-level scoped components

8. Design Guidelines

Typography: Poppins or Inter

Spacing: 4px modular scale

Colors: Food warm tones + futuristic neon

Accessibility: AA contrast standards

9. Development Workflow

Phase 1: UI wireframes + prototype

Phase 2: Login screen development

Phase 3: Auth UI + role routing

Phase 4: Order + Kitchen screens

Phase 5: Billing UI

Phase 6: Testing + polish

Phase 7: Deployment

10. Future Enhancements (Roadmap)

- Offline sync & PWA
- Biometric login
- QR-based ordering
- Loyalty points UI
- Theme switch (light/dark)

- AI-based demand forecasting

11. Testing Standards

UI tests: Cypress / RTL

Accessibility: Axe / Lighthouse

Responsiveness: DevTools / Browserstack

Performance: Lighthouse / WebVitals

12. Delivery Checklist

- Mobile responsive
- Clean reusable components
- Smooth animations
- No console errors
- Accessible and ADA compliant